

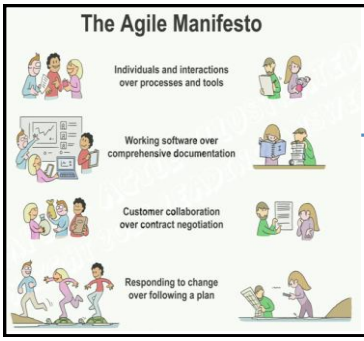
Topic 11

Agile Project management

- Reference Guide - SCRUM & AGILE



- It all started in the spring of ***2000***, when a group of 17 software developers, including Martin Fowler, Jim Highsmith, Jon Kern, Jeff Sutherland, Ken Schwaber, and Bob Martin met in Oregon to discuss how they could speed up **development** times in order bring new software to market faster.
- They recognized two key opportunities that achieving this goal would make possible:
  - Shortening the delay of benefits to users in order to resolve the product-market fit and development graveyard problems
  - Getting feedback from users quickly to confirm the usefulness of new software and continue to improve on it accordingly.



- That is, while there is value in the items on the right, we value the items on the left more.



Comparing Waterfall and Agile Methodologies

Aspect	Waterfall	Agile
Process	Sequential, linear process.	Iterative, incremental process.
Phases	Phases are distinct and do not overlap.	Phases can overlap, allowing for flexibility and adaptability.
Flexibility	Less flexible to changes once development begins.	Embraces changes throughout the project.
Delivery	Entire project is delivered at the end.	Delivers working product incrementally.
Feedback	Feedback is gathered at the end of the project.	Continuous feedback and iteration throughout the project.
Risk	Higher risk of delivering an outdated product.	Risks are managed and addressed throughout the project.
Documentation	Extensive documentation required upfront.	Documentation is light and evolves with the project.
Customer Involvement	Limited customer involvement until the end.	High customer involvement throughout the project.
Adaptability	Less adaptable to changing requirements.	Easily adapts to changing requirements.
Team Structure	Less collaborative, strict roles and responsibilities.	Collaborative, cross-functional teams.
Time to Market	Longer time to market due to sequential nature.	Faster time to market due to iterative development.

Why Agile?

- **Adapting to the Speed of TikTok:**
  - Why wait ages for a product launch when you could be outpaced by a TikTok trend?
- **Surviving the App Store Jungle:**
  - Agile is your survival kit in the wild world of app stores where new apps bloom faster than you can say "download".
- **Netflix Doesn't Wait, Why Should You?:**
  - Agile keeps your product as binge-worthy as the latest Netflix series.
- **No More "That's So 2010":**
  - Agile keeps your product as fresh as the latest viral meme.
- **Customer Feedback, Not Crystal Balls:**
  - Forget crystal balls - Agile gives you real-time feedback from actual users.
- **Avoiding the "Wait, That's Not What I Asked For" Moment:**
  - Agile keeps everyone on the same page, so no more awkward surprises at the end.
- **Life's Too Short for Never-Ending Projects:**
  - Agile means getting stuff done fast and moving on to the next big thing.
- **Because "Fast Fashion" Isn't Just for Clothes Anymore:**
  - In the Agile world, "fast" and "quality" aren't enemies - they're besties.



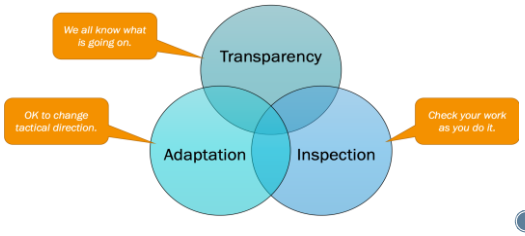
SCRUM

Scrum is an agile way to manage a project, usually software development. Agile software development with Scrum is often perceived as a methodology; but rather than viewing Scrum as methodology, think of it as a framework for managing a process.

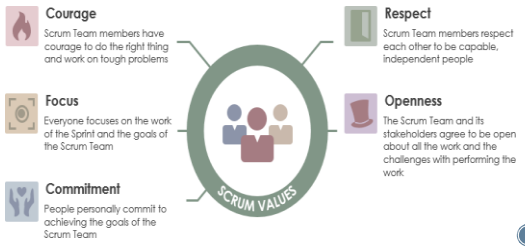
SCRUM THEORY

- Scrum has been around a lot longer than you may think. The first paper on Scrum appeared in the Harvard Business Review in January 1986. Software teams started using the Scrum agile process in 1993.
- Scrum is founded on empiricism and lean thinking.
  - Empiricism asserts that knowledge comes from experience and making decisions based on what is observed.
  - Lean thinking reduces waste and focuses on the essential

SCRUM PILLARS



SCRUM VALUES



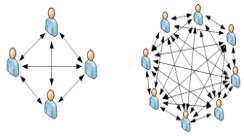


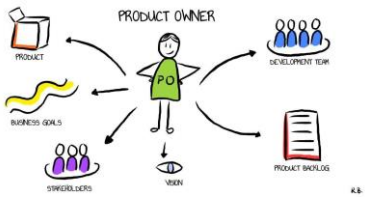
- The fundamental unit of Scrum is a small team of people, a Scrum Team.
- The Scrum Team consists of
  - Scrum Master
  - Product Owner,
  - Developers
- Within a Scrum Team, there are no sub-teams or hierarchies.
- Scrum relies on a self-organizing, cross-functional team.



SCRUM TEAM SIZE

- Scrum Teams are small to stay quick and get a lot done in a Sprint, usually around 10 people or less.
- Smaller teams talk better and work faster.
- If a team gets too big, it's a good idea to split into smaller teams, all working on the same product.
- The best size for a Scrum Team is between 3 to 9 people.





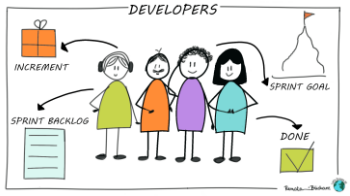
Product Owner Responsibilities:

- **Decide what needs to be done:** The Product Owner decides what features should be built and what order they should be in.
- **Set priorities:** They prioritize the work, deciding which tasks are most important for the team to do first.
- **Talk to everyone:** The Product Owner talks to customers, users, and the team to understand what's needed.
- **Answer questions:** They are the go-to person for the team if they have questions about what to build.
- **Check progress:** They keep an eye on the progress of the project and adjust plans if needed.
- **Accept work:** When the team finishes a task, the Product Owner checks it to make sure it's done right.



- **Scrum Master Responsibilities:**
- **Help the team:**
  - The Scrum Master supports the team in understanding and using Scrum principles and practices.
- **Remove obstacles:**
  - They remove any roadblocks or problems that the team may face, making it easier for them to do their work.
- **Organize meetings:**
  - The Scrum Master arranges and facilitates meetings like Daily Stand-ups to keep the team in sync.
- **Coach the team:**
  - They coach the team on how to work together effectively and improve continuously.
- **Protect the team:** The Scrum Master shields the team from outside distractions, allowing them to focus on their tasks.
- **Ensure Scrum is followed:**
  - They ensure that the team is following the Scrum framework and help them understand its benefits.





Developers Responsibilities:

•**Build the product:** Developers are the ones who actually create the product, writing code, designing interfaces, and making it work.

•**Collaborate:** They work closely with the Product Owner to understand requirements and make sure the product meets customer needs.

•**Work as a team:** Developers help each other out, reviewing each other's code and sharing knowledge.

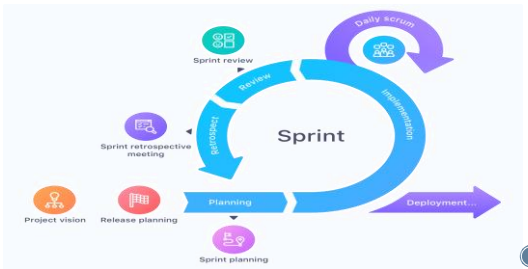
•**Testing:** They write tests to make sure the product works as expected and fix any bugs that come up.

•**Continuous learning:** Developers are always learning new technologies and improving their skills to make better products.

•**Adapt to changes:** They are ready to adapt to changes in requirements or priorities during the project



SCRUM EVENTS



The Sprint



- Sprints: Scrum's Action-Packed Episodes
  - Short, fixed-length events (1 month or less) for consistency.
  - Immediately follow each other for a continuous cycle.
- What Happens:
  - All work to achieve the Product Goal occurs in Sprints.
  - Includes Sprint Planning, Daily Scrums, Sprint Review, and Sprint Retrospective.
- Sprint Rules:
  - No changes that disrupt the Sprint Goal.
  - Quality stays high.
  - Product Backlog can be refined.
  - Scope can be clarified or adjusted as the team learns.
- Summary:
  - Sprints are like Scrum's heartbeat, turning ideas into value in focused, quality bursts.

Sprint Planning

Purpose	Initiates Sprint by planning work to be performed. Collaborative effort of entire Scrum Team.
Duration	Timeboxed to max 8 hours for 1-month Sprint.
Attendees	Scrum Team (Developers, Scrum Master, Product Owner). Other advisors may be invited.
Topic One: Sprint Value	Product Owner proposes how Sprint will increase product value. Scrum Team defines Sprint Goal. Goal finalizedby end of Sprint Planning.
Topic Two: Sprint Scope	Developers, with Product Owner, select Product Backlog items for Sprint. Items may be refined.
Topic Three: How to Do	Developers plan work for each item to meet Definition of Done. Tasks decomposed into smaller items. Decisions made by Developers.
Result: Sprint Backlog	Sprint Goal, selected Product Backlog items, and plan to deliver form Sprint Backlog.

Daily Standup

Purpose	Inspect progress toward Sprint Goal and adjust Sprint Backlog.
Duration	15 minutes
Attendees	Developers of Scrum Team. Product Owner or Scrum Master join if working on Sprint Backlog items.
Frequency	Daily, same time and place during Sprint.
Structure	Developers choose format, focus on Sprint Goal progress, plan next day's work.
Benefits	- Improves communication. Identifies impediments. - Promotes quick decision-making. Reduces need for other meetings.
Adjustments	Developers can adjust plans throughout day, not just in Daily Scrum. Allows for detailed discussions on adapting Sprint work.

Sprint Review

Purpose	Inspect Sprint outcomes, determine adaptations, progress toward Product Goal.
Duration	Timeboxed to max 4 hours for 1-month Sprint.
Attendees	Scrum Team (Developers, Scrum Master, Product Owner). Stakeholders, customers, management.
Process	- Scrum Team presents Sprint results. - Discussion on accomplished work and changes in environment. Collaborative decision-making on next steps.
Product Backlog Adjustments	Based on review, Product Backlog may be adjusted to meet new opportunities.
Working Session	Sprint Review is a working session, not just a presentation.

Sprint Retrospective

Purpose	Plan ways to increase quality and effectiveness.
Duration	Timeboxed to max 3 hours for 1-month Sprint.
Attendees	Scrum Team (Developers, Scrum Master, Product Owner).
Focus Areas	- Inspect last Sprint: individuals, interactions, processes, tools, Definition of Done. - Identify assumptions that led to issues.
Discussion Points	- What went well in the Sprint? - Problems encountered and their solutions (or lack thereof).
Improvement Identification	Scrum Team identifies changes to improve effectiveness, focusing on most impactful improvements.
Implementation	Most impactful improvements are addressed ASAP, may be added to next Sprint Backlog.

SCRUM ARTIFACTS

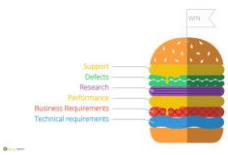


Scrum's artifacts are like the key tools that show the work or value being done. They're all about making sure everyone knows what's going on. When everyone looks at these artifacts, they're on the same page and can make decisions together.

Here's what each artifact is committed to showing:

- **Product Backlog:**
  - It's all about the "Product Goal" - this keeps the team focused on what they want to achieve.
- **Sprint Backlog:**
  - Here, it's the "Sprint Goal" - this keeps the team on track for the current Sprint.
- **Increment:**
  - This is where the "Definition of Done" comes in - it sets the standard for when something is truly finished.

Product Backlog Structure

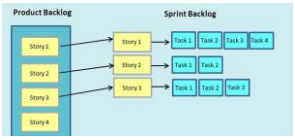


- **Product Backlog: What It Is**
  - It's a dynamic list of what's needed to make the product better.
  - This list guides all the work the Scrum Team does.
- **Ready for Sprint Planning**
  - Items that can be finished in one Sprint are ready for Sprint Planning.
  - They get this clear after refining activities in Product Backlog refinement.
- **Product Backlog Refinement**
  - Breaking down items into smaller, detailed tasks.
  - Adds descriptions, priorities, and sizes to items.
  - This ongoing process helps make planning easier.
- **Responsibility for Sizing**
  - Developers decide on the size of tasks they can handle.
  - Product Owner can guide with trade-offs and understanding.
- **Commitment: Product Goal**
  - The Product Goal is like the target for the team.
  - It's in the Product Backlog and guides everything else in it.
  - Product Backlog items emerge to fulfill this Product Goal.

- What's a Product?
  - A product is the thing that delivers value.
  - It's clear what it is, who it's for, and its purpose.
  - This could be a service, a physical product, or something else.
- Product Goal: The Long-Term Objective
  - The Product Goal is the big picture, the long-term aim for the product.
  - It's what the team plans against and aims to achieve over time.

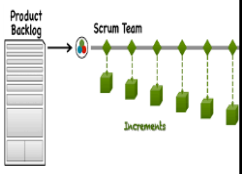
Sprint Backlog

- **What's in the Sprint Backlog:**
  - The **Sprint Goal** (why),
  - Selected **Product Backlog items** for the Sprint (what),
  - An actionable **plan** for how to deliver the Increment (how).
- **Plan for Developers, by Developers:**
  - The Sprint Backlog is all about the Developers.
  - It's their plan to achieve the Sprint Goal.
  - It's visible, real-time, and changes as the team learns more.
- **Dynamic and Detailed:**
  - It's constantly updated throughout the Sprint.
  - Should be detailed enough for Developers to track progress in Daily Scrum.



Increment

- An Increment is a concrete stepping stone toward the Product Goal.
- Each Increment is additive to all prior Increments and thoroughly verified, ensuring that all Increments work together. In order to provide value, the Increment must be usable





•Commitment: Sprint Goal	•Creating the Sprint Goal:	•Increment: Stepping Stone to Product Goal	•Delivery of Increment:	•Definition of Done: Quality Standard	•Enforcing Definition of Done:
• The Sprint Goal is the single objective for the Sprint. • It's a commitment by the Developers but allows flexibility in how to achieve it. • Creates coherence and focus, encouraging teamwork on one goal.	• Made during Sprint Planning and added to Sprint Backlog. • Developers keep it in mind throughout the Sprint. • If work changes, they collaborate with Product Owner to adjust without changing the Sprint Goal.	• Each Increment is a step toward the Product Goal. • They're added up and thoroughly checked to ensure they work together. • To add value, the Increment must be usable. • Multiple Increments can be made in a Sprint, supporting empiricism.	• Increments can be presented at Sprint Review, supporting transparency. • Increment may be delivered to stakeholders before Sprint ends, not waiting for Sprint Review.	• It's a clear description of what's needed for an Increment to be considered done. • When a Product Backlog item meets this, it becomes an Increment. • Defines transparency by showing what work is completed. • If something doesn't meet this standard, it goes back to Product Backlog.	• All Scrum Teams must follow the Definition of Done, whether it's an organizational standard or one created by the team. • Developers must adhere to this standard. • If multiple Scrum Teams work together, they must agree on and follow the same Definition of Done.

END OF TOPIC 11

COMING UP!!!!!!
- User Stories
- Communication & Quality
- Concluding Topics
- Semester Exam
- Project Submission

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